

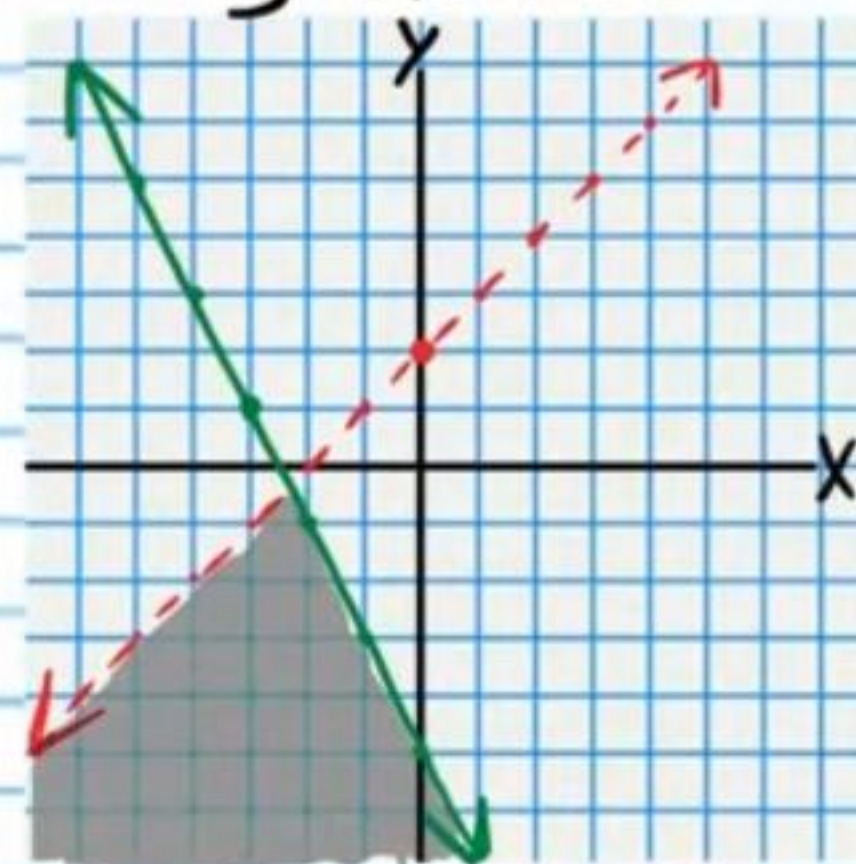
Practice Intermediate Algebra Test I (Ver 1)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- II) Show all work.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

① Show the solutions of the following system of inequalities by graphing.

$$\begin{cases} y < x + 2 \\ y - 1 \leq -2(x + 3) \end{cases}$$



② Determine if the following equation has one solution, no solution, or infinite solutions. Prove algebraically.

$$-5x + 1 = 2x - 7x + 1$$

$$-5x + 1 = -5x + 1$$

$$1 = 1$$

Infinite Solutions

③ Solve.

$$2\left|\frac{x}{3} + 1\right| - 4 = -4$$

$$2\left|\frac{x}{3} + 1\right| = 0$$

$$\left|\frac{x}{3} + 1\right| = 0$$

$$\frac{x}{3} + 1 = -0 \text{ or } \frac{x}{3} + 1 = 0$$

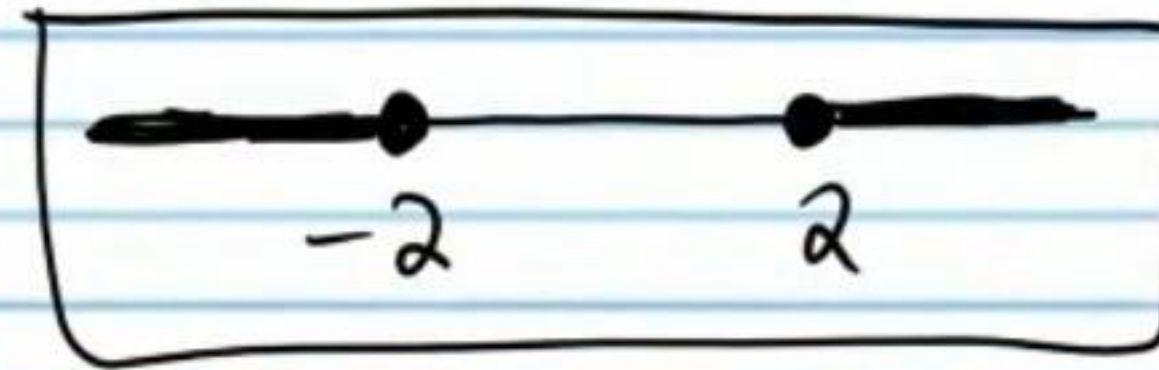
$$x = -3$$

④ Find the algebraic solution and the graphical solution of the following inequality.

$$|x| + 2 \geq 4$$

$$|x| \geq 2$$

$$x \leq -2 \text{ or } x \geq 2$$



⑤ Simplify. *Do not assume that x is positive.*

$$\sqrt[6]{x^{30}} = |x^5|$$

⑥ Simplify.

$$979 = -1$$

$$\begin{array}{r} 19 \\ 4 \overline{) 79} \\ \underline{4} \\ 39 \\ \underline{36} \\ 3 \end{array}$$

⑦ Solve.

$$\begin{cases} 3x + 2y - z = -10 \\ x = 2y + 3z - 14 \\ x - y - z = -6 \end{cases}$$

$$\rightarrow 3(2y + 3z - 14) + 2y - z = -10$$

$$6y + 9z - 42 + 2y - z = -10$$

$$8y + 8z = 32$$

$$y + z = 4$$

$$2y + 3z - 14 - y - z = -6$$

$$y + 2z = 8$$

$$\begin{cases} y + z = 4 \\ y + 2z = 8 \end{cases}$$

$$0 - z = -4$$

$$-z = -4$$

$$z = 4$$

$$y + 4 = 4$$

$$y = 0$$

$$x = 2(0) + 3(4) - 14$$

$$x = -2$$

$$(-2, 0, 4)$$

Simplify and write in $a + bi$ form.

$$\textcircled{8} -5i(2i + 7)$$

$$-10i^2 - 35i$$

$$-10(-1) - 35i$$

$$10 - 35i$$

$$\textcircled{9} (2i + 6) - (i - 7)$$

$$2i + 6 - i + 7$$

$$13 + i$$

$$\textcircled{10} \frac{i}{4i - 2}$$

$$\frac{i}{(4i - 2)(4i + 2)}$$

$$\frac{4i^2 + 2i}{16i^2 + 8i - 8i - 4}$$

$$\frac{4(-1) + 2i}{16(-1) - 4}$$

$$\frac{-4 + 2i}{-16 - 4}$$

$$\frac{-4 + 2i}{-20} = \frac{-4}{-20} + \frac{2i}{-20} = \frac{1}{5} - \frac{1}{10}i$$

⑪ Rationalize the denominator.

$$\frac{\sqrt{5}-1}{\sqrt{2}+3}$$

$$\frac{(\sqrt{5}-1)(\sqrt{2}-3)}{(\sqrt{2}+3)(\sqrt{2}-3)}$$

$$\frac{\sqrt{10}-3\sqrt{5}-\sqrt{2}+3}{\sqrt{4}-3\sqrt{2}+3\sqrt{2}-9}$$

$$\frac{\sqrt{10}-3\sqrt{5}-\sqrt{2}+3}{2-9}$$

$$\boxed{\frac{\sqrt{10}-3\sqrt{5}-\sqrt{2}+3}{-7}}$$

⑫ Simplify. Assume that x is positive.

$$\frac{\sqrt{50x^3}}{\sqrt{2x}}$$

$$\sqrt{\frac{50x^3}{2x}}$$

$$\sqrt{25x^2}$$

$$\boxed{5x}$$

⑬ Simplify. Assume that x is positive.

$$8x^3\sqrt{9x} + 2\sqrt{81x^7}$$

$$8x^3 \cdot \sqrt{9} \cdot \sqrt{x} + 2\sqrt{81} \sqrt{x^7}$$

$$8x^3 \cdot 3 \cdot \sqrt{x} + 2 \cdot 9 \cdot \sqrt{x^7}$$

$$24x^3\sqrt{x} + 18\sqrt{x^7}$$

$$24x^3\sqrt{x} + 18\sqrt{x^6 \cdot x}$$

$$24x^3\sqrt{x} + 18\sqrt{x^6} \cdot \sqrt{x}$$

$$24x^3\sqrt{x} + 18 \cdot x^3 \cdot \sqrt{x}$$

$$24x^3\sqrt{x} + 18x^3\sqrt{x}$$

$$\boxed{42x^3\sqrt{x}}$$

⑭ Solve by completing the square.

$$0 = x^2 + 10x + 9$$

$$-9 = x^2 + 10x$$

$$25 - 9 = x^2 + 10x + 25$$

$$16 = (x+5)(x+5)$$

$$16 = (x+5)^2$$

$$\pm\sqrt{16} = \sqrt{(x+5)^2}$$

$$\pm 4 = x+5$$

$$\pm 4 - 5 = x$$

$$\begin{array}{l} \swarrow \quad \searrow \\ x = 4 - 5 \quad \text{or} \quad x = -4 - 5 \end{array}$$

$$\boxed{x = -1 \quad \text{or} \quad x = -9}$$

⑮ Solve by completing the square.

$$2x^2 - 28x + 96 = 0$$

$$\frac{2x^2 - 28x + 96}{2} = \frac{0}{2}$$

$$x^2 - 14x + 48 = 0$$

$$x^2 - 14x = -48$$

$$x^2 - 14x + 49 = -48 + 49$$

$$x^2 - 14x + 49 = 1$$

$$(x-7)(x-7) = 1$$

$$(x-7)^2 = 1$$

$$\sqrt{(x-7)^2} = \pm\sqrt{1}$$

$$x-7 = \pm 1$$

$$x = 7 \pm 1$$

$$x = 7+1 \text{ or } x = 7-1$$

$$\boxed{x=8 \text{ or } x=6}$$

⑯ Find the discriminant of the following equation and use it to determine the number of and type of solutions. Write "one real," "two real," or "two complex."

$$3x^2 + 5x - 10 = 0$$

$$a=3, b=5, c=-10$$

$$D = b^2 - 4ac$$

$$D = 5^2 - 4(3)(-10)$$

$$D = 25 + 120$$

$$D = 145$$

Two real solutions

⑰ Solve.

$$\begin{cases} a & 2x + 5y + z = -10 \\ b & -x - 2y + 2z = 6 \\ c & 4x - y - 3z = 8 \end{cases}$$

$$\begin{cases} b & -x - 2y + 2z = 6 \\ -2c & -8x + 2y + 6z = -16 \end{cases}$$

$$-9x + 8z = -10$$

$$\begin{cases} a & 2x + 5y + z = -10 \\ 5c & 20x - 5y - 15z = 40 \end{cases}$$

$$22x - 14z = 30$$

$$\frac{a+5c}{2} \Rightarrow 11x - 7z = 15$$

$$\begin{cases} d & -9x + 8z = -10 \\ e & 11x - 7z = 15 \end{cases}$$

$$-9x + 8(1) = -10$$

$$-9x + 8 = -10$$

$$-9x = -18$$

$$\boxed{x=2}$$

$$2(2) + 5y + 1 = -10$$

$$4 + 5y + 1 = -10$$

$$\boxed{y=-3}$$

$$11d \begin{cases} -99x + 88z = -110 \\ 9c & 99x - 63z = 135 \end{cases}$$

$$25Z = 25$$

$$25Z = 25$$

$$\boxed{Z=1}$$

$$\boxed{(2, -3, 1)}$$

⑮ Solve by completing the square.

$$0 = x^2 - 3x + 4$$

$$-4 = x^2 - 3x$$

$$-4 + \frac{9}{4} = x^2 - 3x + \frac{9}{4}$$

$$-\frac{7}{4} = (x - \frac{3}{2})(x - \frac{3}{2})$$

$$-\frac{7}{4} = (x - \frac{3}{2})^2$$

$$\pm\sqrt{-\frac{7}{4}} = \sqrt{(x - \frac{3}{2})^2}$$

$$\pm\sqrt{-\frac{7}{4}} = x - \frac{3}{2}$$

$$\pm\sqrt{-\frac{7}{4}} + \frac{3}{2} = x$$

$$\pm i\sqrt{\frac{7}{4}} + \frac{3}{2} = x$$

$$\pm i\frac{\sqrt{7}}{\sqrt{4}} + \frac{3}{2} = x$$

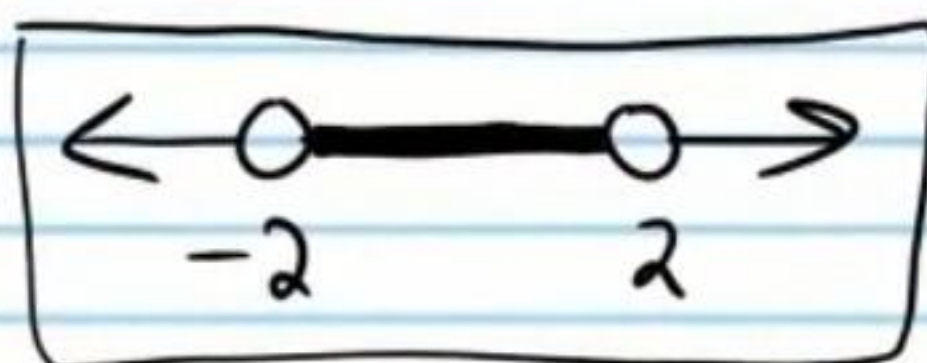
$$\pm \frac{i\sqrt{7}}{2} + \frac{3}{2} = x$$

⑯ Find the algebraic solution and the graphical solution of the following inequality.

$$|5x| < 10$$

$$5x > -10 \text{ and } 5x < 10$$

$$x > -2 \text{ and } x < 2$$



⑰ Solve by completing the square.

$$5x^2 - 3x + 8 = 0$$

$$\frac{5x^2 - 3x + 8}{5} = \frac{0}{5}$$

$$x^2 - \frac{3}{5}x + \frac{8}{5} = 0$$

$$x^2 - \frac{3}{5}x = -\frac{8}{5}$$

$$x^2 - \frac{3}{5}x + \frac{9}{100} = -\frac{8}{5} + \frac{9}{100}$$

$$(x - \frac{3}{10})(x - \frac{3}{10}) = \frac{-160}{100} + \frac{9}{100}$$

$$(x - \frac{3}{10})^2 = \frac{-151}{100}$$

$$\sqrt{(x - \frac{3}{10})^2} = \pm\sqrt{\frac{-151}{100}}$$

$$x - \frac{3}{10} = \pm i\frac{\sqrt{151}}{10}$$

$$x = \frac{3}{10} \pm \frac{i\sqrt{151}}{10}$$

⑱ Solve using the quadratic formula.

$$0 = -x^2 + x + 5$$

$$a = -1, b = 1, c = 5$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-1 \pm \sqrt{1^2 - 4(-1)(5)}}{2(-1)}$$

$$x = \frac{-1 \pm \sqrt{1 + 20}}{-2}$$

$$x = \frac{-1 \pm \sqrt{21}}{-2}$$

$$x = \frac{1 \pm \sqrt{21}}{2}$$

②② Solve using the quadratic formula.

$$4x^2 - 6x + 9 = 0$$

$$a=4, b=-6, c=9$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(4)(9)}}{2(4)}$$

$$x = \frac{6 \pm \sqrt{36 - 144}}{8}$$

$$x = \frac{6 \pm \sqrt{-108}}{8}$$

$$x = \frac{6 \pm i\sqrt{108}}{8}$$

$$x = \frac{6 \pm i6\sqrt{3}}{8}$$

$$x = \frac{3 \pm 3i\sqrt{3}}{4}$$

②③ Simplify and rationalize the denominator. Assume that x is positive.

$$\sqrt{\frac{20}{45x^5}}$$

$$\frac{\sqrt{20}}{\sqrt{45x^5}}$$

$$\frac{\sqrt{4 \cdot 5}}{\sqrt{9 \cdot 5 \cdot x^4 \cdot x}}$$

$$\frac{\sqrt{4} \cdot \sqrt{5}}{\sqrt{9} \cdot \sqrt{5} \cdot \sqrt{x^4} \cdot \sqrt{x}}$$

$$\frac{2\sqrt{5}}{3\sqrt{5} x^2 \sqrt{x}}$$

$$\frac{2\sqrt{5}}{3x^2 \sqrt{5x}}$$

$$\frac{2\sqrt{5} \cdot \sqrt{5x}}{3x^2 \sqrt{5x} \cdot \sqrt{5x}}$$

$$\frac{2\sqrt{25x}}{3x^2 \sqrt{25x^2}} = \frac{2\sqrt{25x}}{3x^2(5x)} = \frac{2\sqrt{25x}}{15x^3}$$

②④ Simplify and rationalize the denominator. Assume that x is a positive number.

$$\frac{2}{\sqrt[3]{81x^5}} \quad \sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}$$

$$\frac{2}{\sqrt[3]{27 \cdot 3 \cdot x^3 \cdot x^2}}$$

$$\frac{2}{\sqrt[3]{27} \cdot \sqrt[3]{3} \cdot \sqrt[3]{x^3} \cdot \sqrt[3]{x^2}}$$

$$\frac{2}{3 \cdot \sqrt[3]{3} \cdot x \cdot \sqrt[3]{x^2}}$$

$$\frac{2}{3x \sqrt[3]{3x^2}}$$

$$\frac{2}{3x \sqrt[3]{3x^2}} \cdot \frac{\sqrt[3]{9x}}{\sqrt[3]{9x}}$$

$$\frac{2\sqrt[3]{9x}}{3x \sqrt[3]{27x^3}}$$

$$\frac{2\sqrt[3]{9x}}{3x(3x)} = \frac{2\sqrt[3]{9x}}{9x^2}$$

②⑤ Solve.

$$\frac{|-2x+1|}{-4} + 1 \geq 2$$

$$\frac{|-2x+1|}{-4} \geq 1$$

$$|-2x+1| \leq -4$$

No Solution